Koneru Lakshmaiah Education Foundation
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MACHINE LEARNING (25CS2107E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-5	Team No- 64	Project Title-Wine Quality Prediction and Feature Importance Analysis Using Random Forest

1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)

The determination of the quality of wine has been a challenging issue as it is performed through expert sensory panels, which are subjective and hard to automate. The proposed work attempts to determine wine quality through a supervised machine learning approach by applying a Random Forest algorithm and determining the importance of physicochemical factors (alcohol, volatile acidity, citric acid, sulphates, pH) for predicting the wine quality from the UCI Wine Quality dataset.

Apart from the predictive performance of the model, the importance of each factor will be analyzed based on the Gini importance measure, and the results will be validated against scientific knowledge about wine production. The performance of the classification model will be measured based on accuracy, precision, recall, F1-score, and a confusion matrix using k-fold cross-validation.

Thus, by combining the predictive performance of the model with its explainability and feature level importance measures, the research illustrates how ensemble learning approaches could become more than just black box models in the context of practical tasks.

2. Team Details

Roll Number(s) Name of the Student	Signature
2520030305 Sura.Rahul Reddy	
2520030506 Yashwanth	

3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

- Python 3
- Pandas, NumPy – data handling and preprocessing
- Scikit-learn – Random Forest Classifier, train-test split, cross-validation, evaluation metrics
- Matplotlib, Seaborn – data visualization and feature importance plotting
- Jupyter Notebook, Google Colab – development environment
- Kaggle – dataset source: Wine Quality Dataset, UCI Machine Learning Repository

Guide Name and Signature (remarks if any) _____